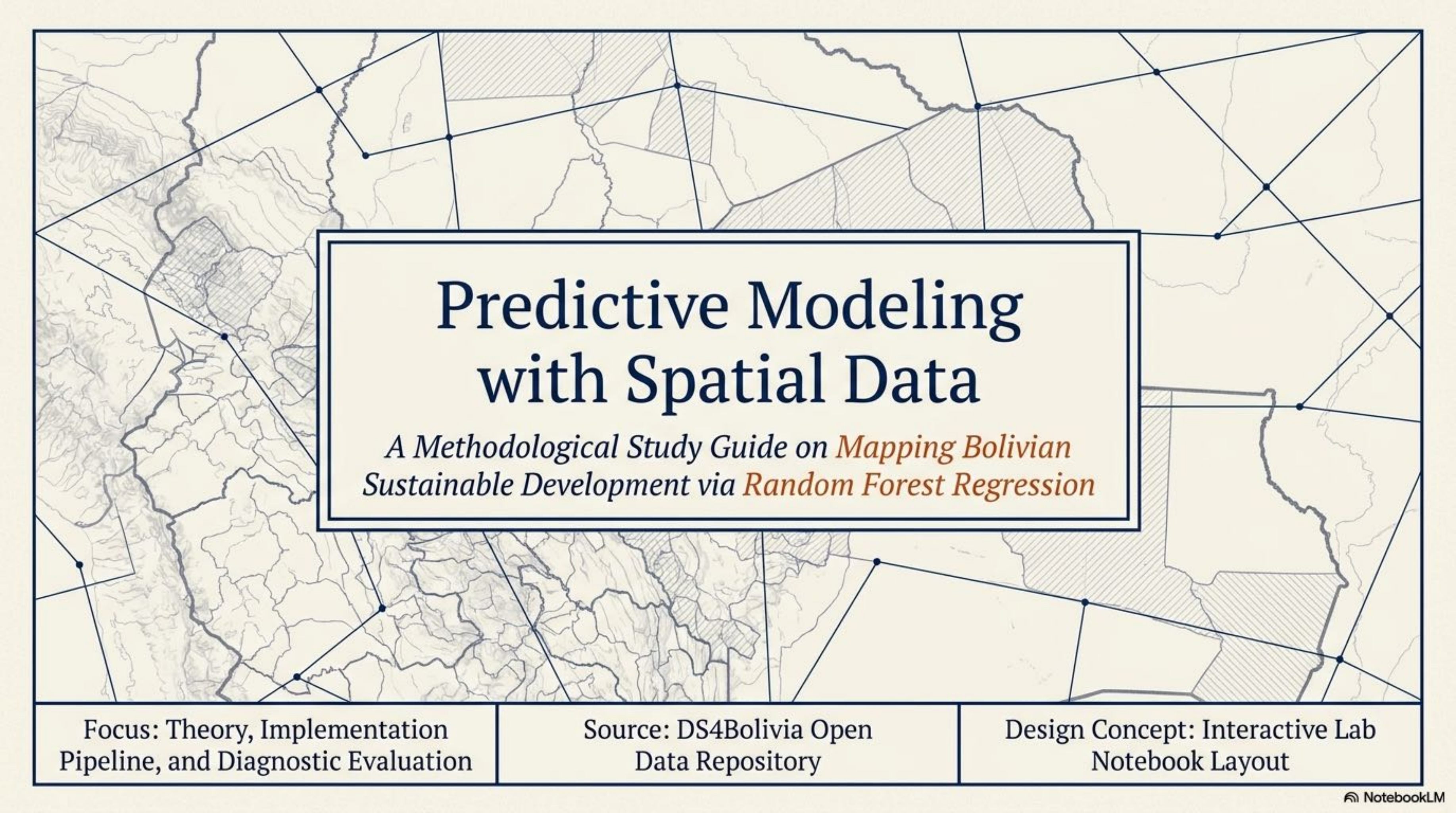




Predictive Modeling with Spatial Data

*A Methodological Study Guide on **Mapping Bolivian**
Sustainable Development via **Random Forest Regression***

Focus: Theory, Implementation
Pipeline, and Diagnostic Evaluation

Source: DS4Bolivia Open
Data Repository

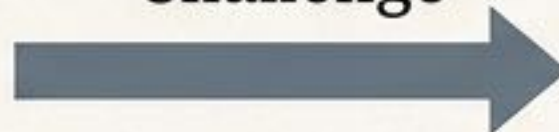
Design Concept: Interactive Lab
Notebook Layout

Traditional Survey Data



Slow, Expensive, Sparse

The Analytical
Challenge



Satellite Proxy Data



Fast, Low-Cost, **Complete Coverage**

The Objective

Predict Bolivia's Municipal Sustainable Development Index (IMDS) using machine learning to evaluate if satellite imagery contains genuine socioeconomic signals.

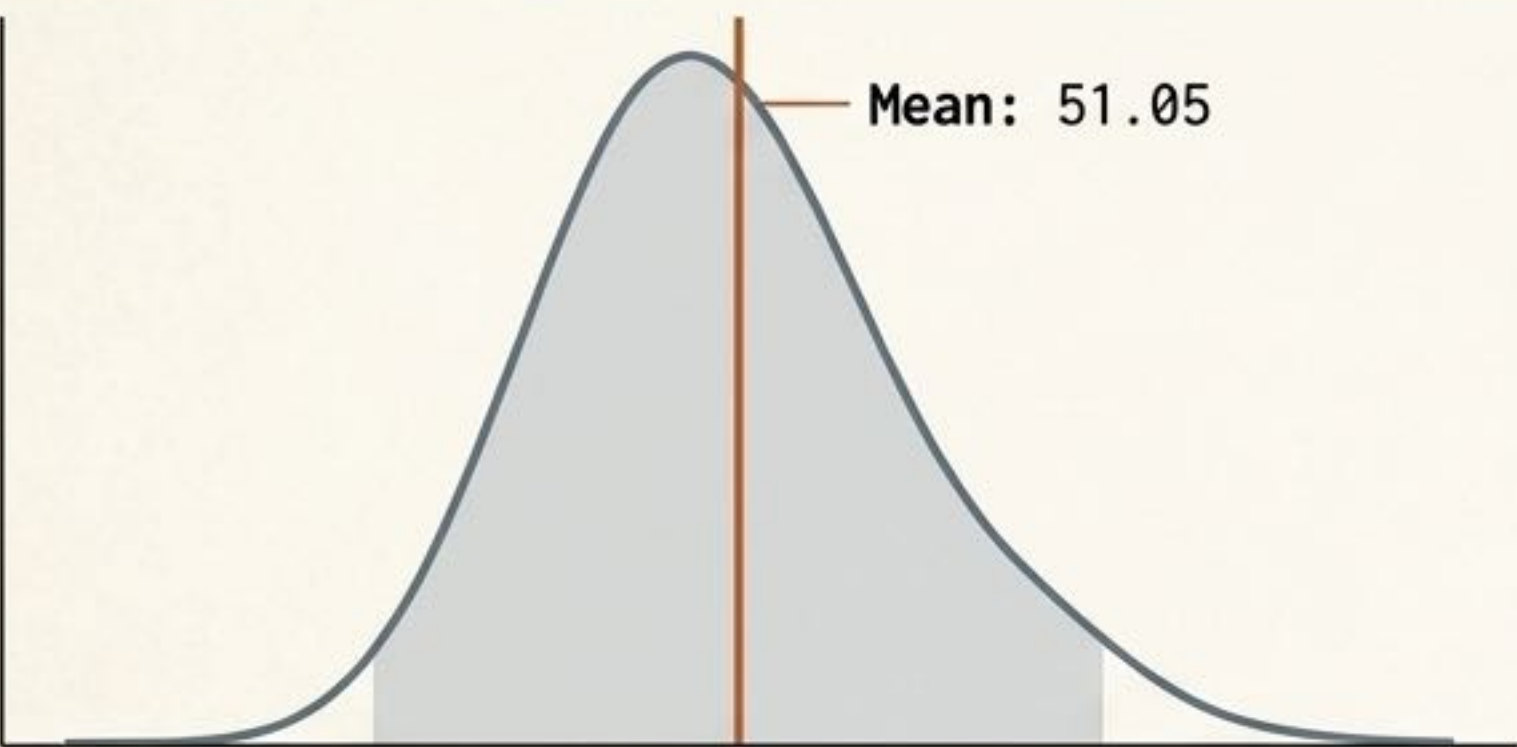
Analytical Hurdles

1. Extracting non-linear signals.

2. Managing high-dimensional, multicollinear spatial data without overfitting.

The Target Variable (Y)

Municipal Sustainable Development Index (IMDS)



Distribution: 0-100 scale
Coverage: N = 339 municipalities (100%)
Standard Deviation: 6.77

Diagnostic Note

The dense clustering of scores between 47-55 heavily influences predictive bias at extreme tails.

The Feature Space (X)

Satellite Embeddings

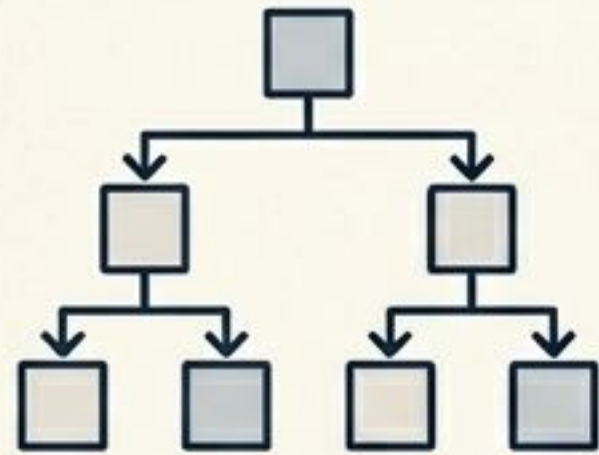
A00	A12	A12	A41	A63	...	A63
A00	A12	A12	A41	A63	...	A63
...	...	...	...	...	...	...
A00	A12	A12	A41	A63	...	A63
A00	A12	A12	A41	A63	...	A63
A00	A12	A12	A41	A63	...	A63
...	...	...	...	...	...	...
A00	A12	A12	A41	A63	...	A63

64-
Dimensional
Vectors

Data Source: Extracted visual information (land use, urbanization, terrain) from 2017 satellite imagery, mathematically compressed.

Theoretical Foundations: The Forest Algorithm

Decision Tree

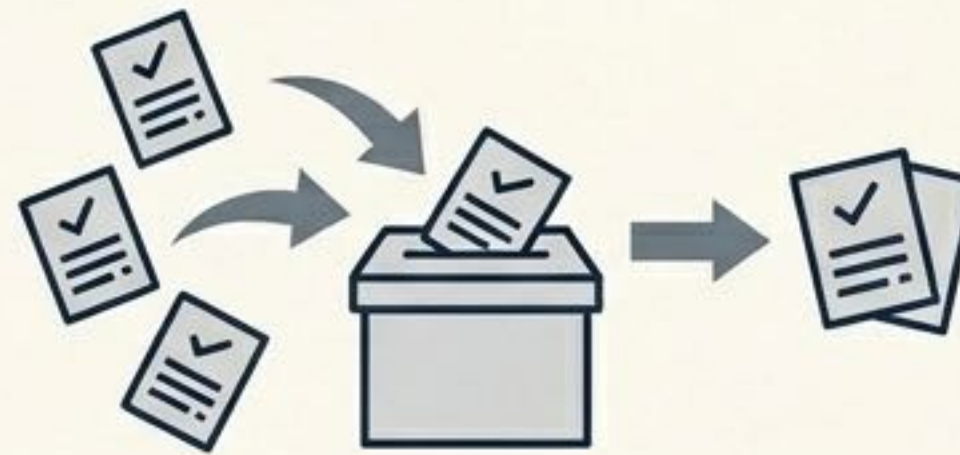


Definition: Recursive binary splits on features.

Analogy: A flowchart of yes/no verdicts.

Example: Splitting first on embedding A12 (urban signal), then A41 (vegetation index).

Bagging

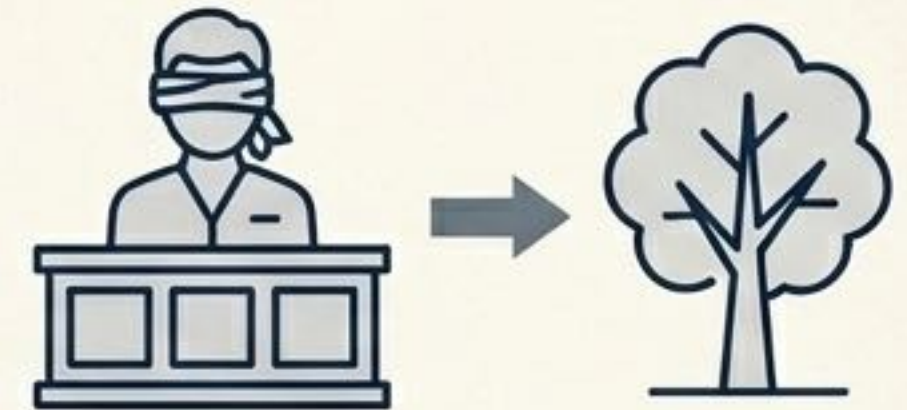


Definition: Averaging predictions of B trees on resampled data to reduce variance.

Analogy: Polling diverse juries and averaging their verdicts.

Example: Averaging predictions from 500 different bootstrap trees.

Random Forest



Definition: Bagging + **random feature subsets** at each split to decorrelate trees.

Analogy: Blindfolding jurors to random subsets of evidence.

Example: Considering only $\sqrt{64} = 8$ **dimensions** per split.

The Implementation Pipeline

1.

Pre-Modeling

Exploratory Data
Analysis (EDA) &

The Isolation
Principle
(Data Splitting)

2.

**Baseline
Model**

Algorithm
Initialization
& Honest
Evaluation
via Cross-
Validation

3.

**Hyperparameter
Tuning**

Algorithmic
Constraint
Optimization
via Randomized
Search

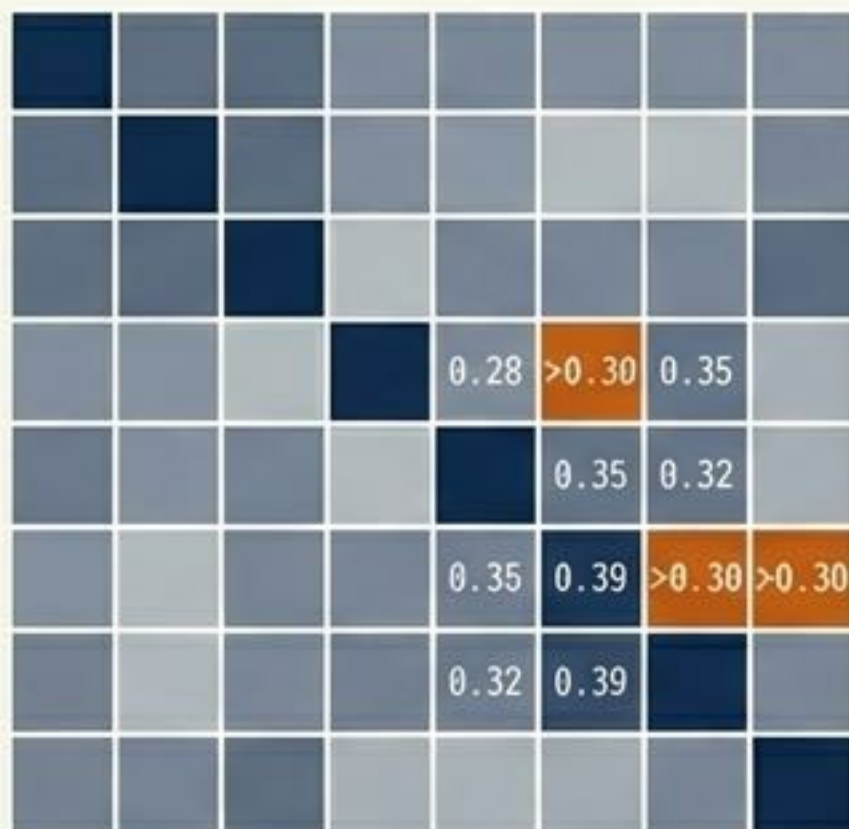
4.

**Empirical
Evaluation**

Out-of-Sample
Testing,
Interpretability
Metrics, and
Economic
Diagnostics

Step 1: Pre-Modeling & The Isolation Principle

Multicollinearity Management



Observation: Moderate correlations (0.25–0.40) exist between **embeddings** and the **IMDS** target.

Handling: Multicollinearity is naturally managed by the Random Forest via **random feature subsets** at each split.

The Isolation Principle (Train/Test Split)

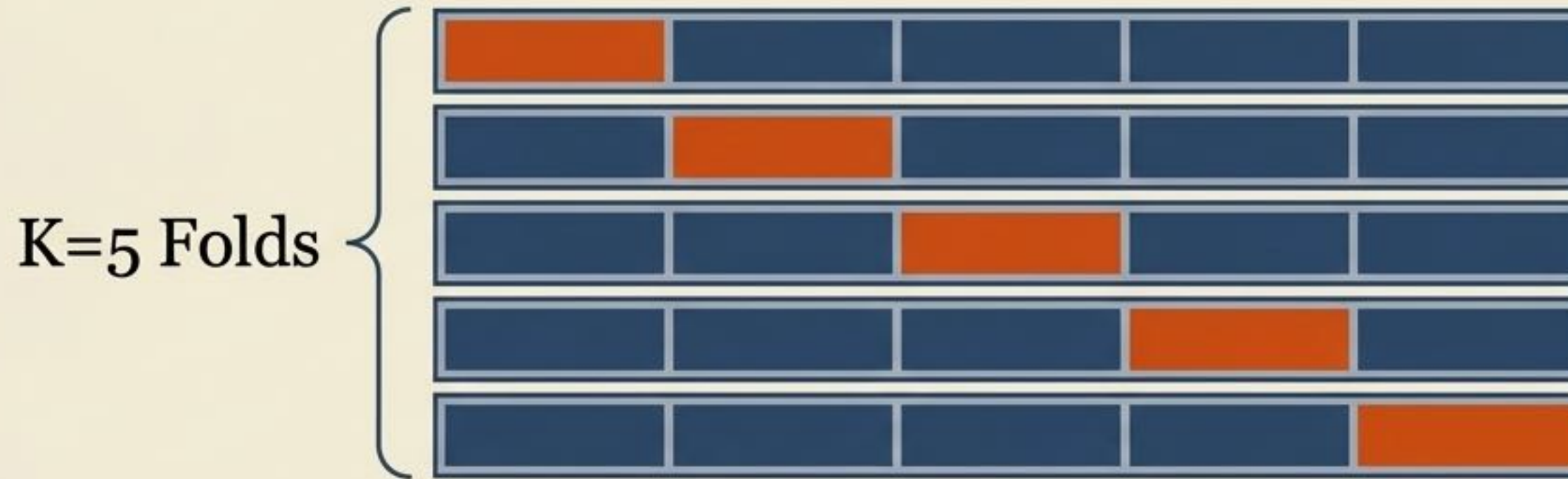


Definition: Isolating **D_{train}** from **D_{test}** to prevent data leakage.

Analogy: Practicing on yesterday's exam, graded on tomorrow's.

Implementation: **80/20** split. Training (N=271), Test (N=68). The test set represents **~1.5%** of data per prediction, necessitating robust validation.

Step 2: Baseline Model & Honest Validation



Concept Framework

Definition: Splitting the training set into K folds. Train on K-1, validate on 1. Rotate to stabilize estimates without touching the test set.

Analogy: Taking five mock exams and averaging the scores for a steadier read.

Implementation & Results

Goal: Establish a performance floor using default scikit-learn parameters.

Result: 5-fold CV Mean $R^2 = 0.2526$
Standard Deviation: ± 0.0728

Insight: High variability across folds reflects the small sample size (N=271 train), proving the necessity of K-fold validation.

Step 3: Hyperparameter Tuning

Method: RandomizedSearchCV

```
Search_Space = {  
    n_estimators: [100, 300, 500],  
    max_depth: [10, 20, 30, None],  
    max_features: ['sqrt', 'log2']  
}
```

```
Optimal_Config = {  
    n_estimators = 500,  
    max_depth = 30,  
    max_features = 'sqrt'  
}
```

Concept Framework

Definition: Searching over a grid of algorithmic constraints to optimize the CV score.

Analogy: Adjusting oven knobs (temperature, time, rack placement) before baking the final cake.

The Result

Best CV $R^2 = 0.2721$

Insight: The optimized configuration yields a **minimal 2-point improvement** over the baseline. The algorithm has stabilized.

Step 4: Empirical Evaluation

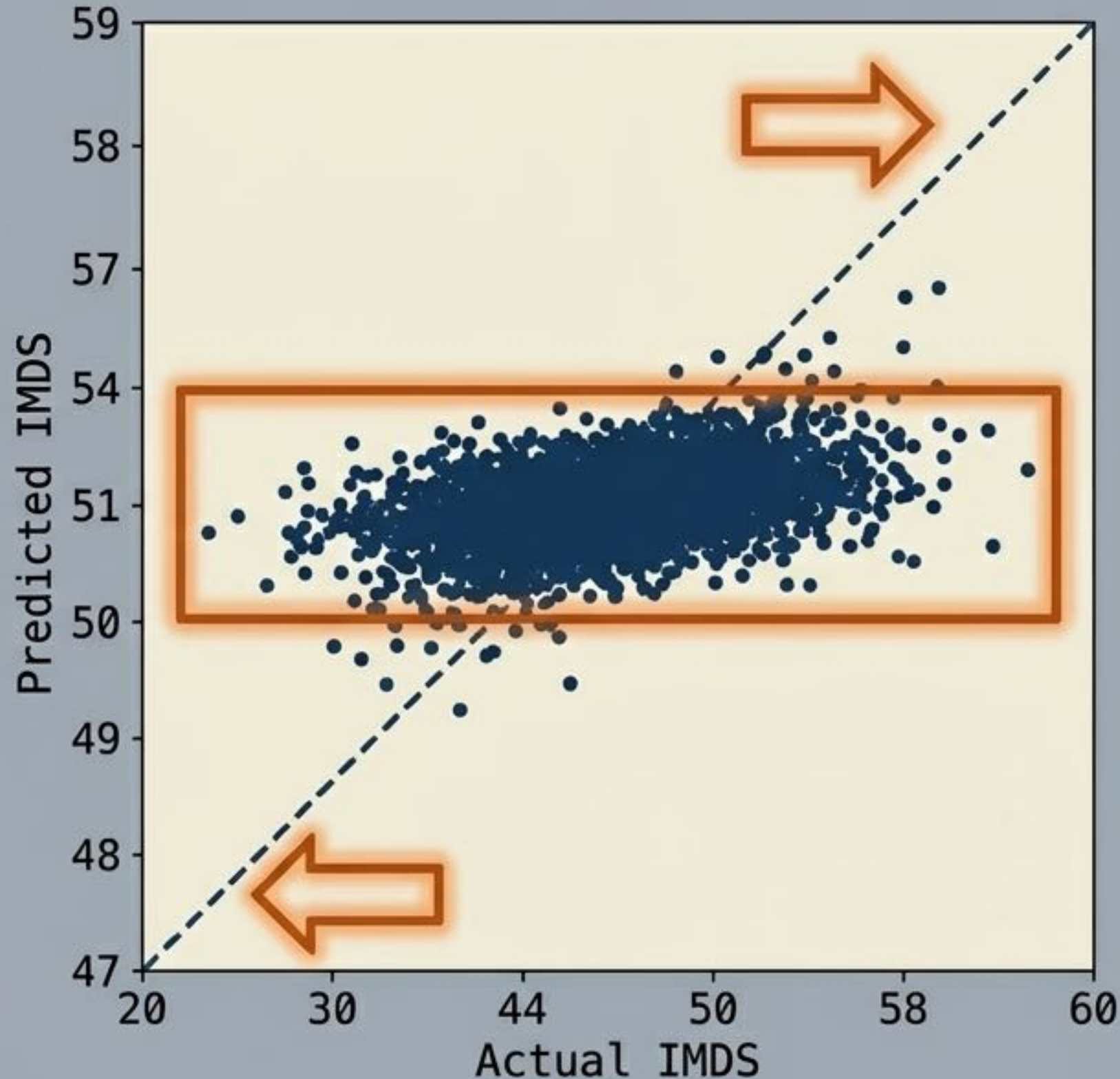
Metric	Baseline Default	Tuned Optimized
R ²	0.2307	0.2297
RMSE	6.52	6.52
MAE	4.68	4.72

Diagnostic Insight: The Performance Ceiling

- Observation: Sophisticated hyperparameter tuning yielded negligible out-of-sample improvement.
- Conclusion: The predictive ceiling is dictated by the features themselves, not model configuration. The final model successfully explains ~23% of IMDS variation on out-of-sample data.

$$R^2 = 1 - \left(\frac{SS_{\text{res}}}{SS_{\text{tot}}} \right)$$
$$RMSE = \sqrt{\text{mean}(\text{errors}^2)}$$
$$MAE = \text{mean}(|\text{errors}|)$$

Diagnostic Analysis: Out-of-Sample Performance



Diagnostic 1: Regression to the Mean

Predictions tightly cluster in the 47-55 range, near the training mean. The model pulls extreme values toward the center.

Diagnostic 2: Heteroscedasticity

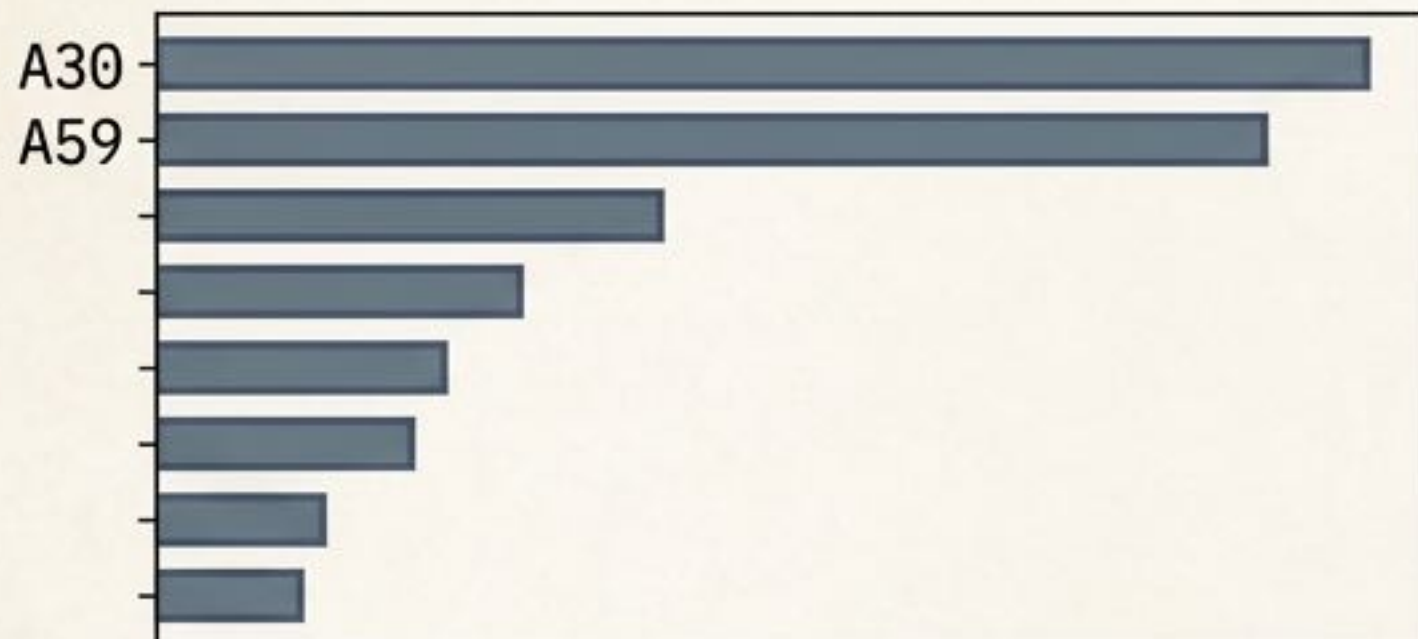
Residual variance is not constant; errors spread significantly wider at the extremes (the highest and lowest IMDS municipalities).

Economic Policy Implication

The model cannot reliably identify the absolute highest- or lowest-performing municipalities. Policymakers using this solely for targeted aid would miss the most vulnerable (extreme) cases.

Interpretability Matrix: Feature Importance Methodologies

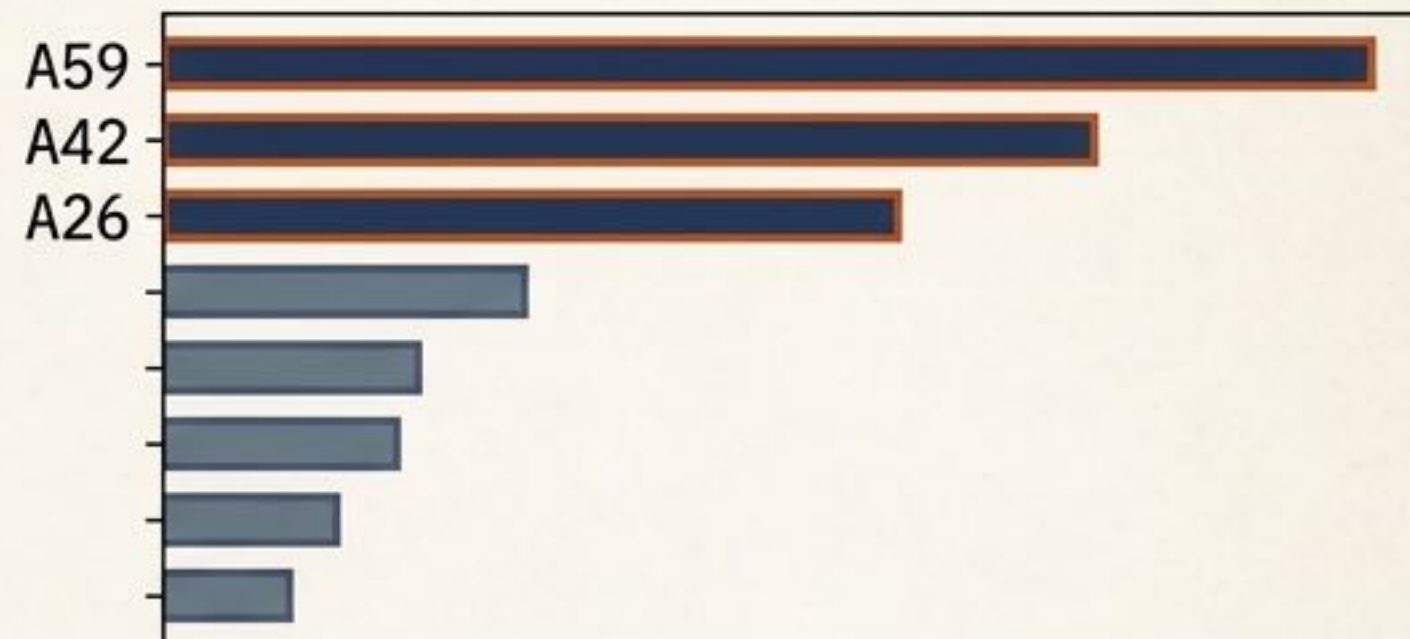
Method 1: Mean Decrease in Impurity (MDI)



Mechanism: Sums impurity reduction across all splits.

Bias: Inherently inflates the importance of continuous or high-cardinality features.

Method 2: Permutation Importance



Mechanism: Randomly scrambles a feature in the test set to measure the exact drop in R^2 .

Advantage: Unbiased by scale; evaluates true out-of-sample contribution.

The True Signal Extract

Permutation reveals dimension **A59** as the dominant predictor, followed closely by **A42** and **A26**. **A26**. The satellite signal is distributed across multiple spatial patterns, not monolithic.

Understanding Non-Linearity: Partial Dependence

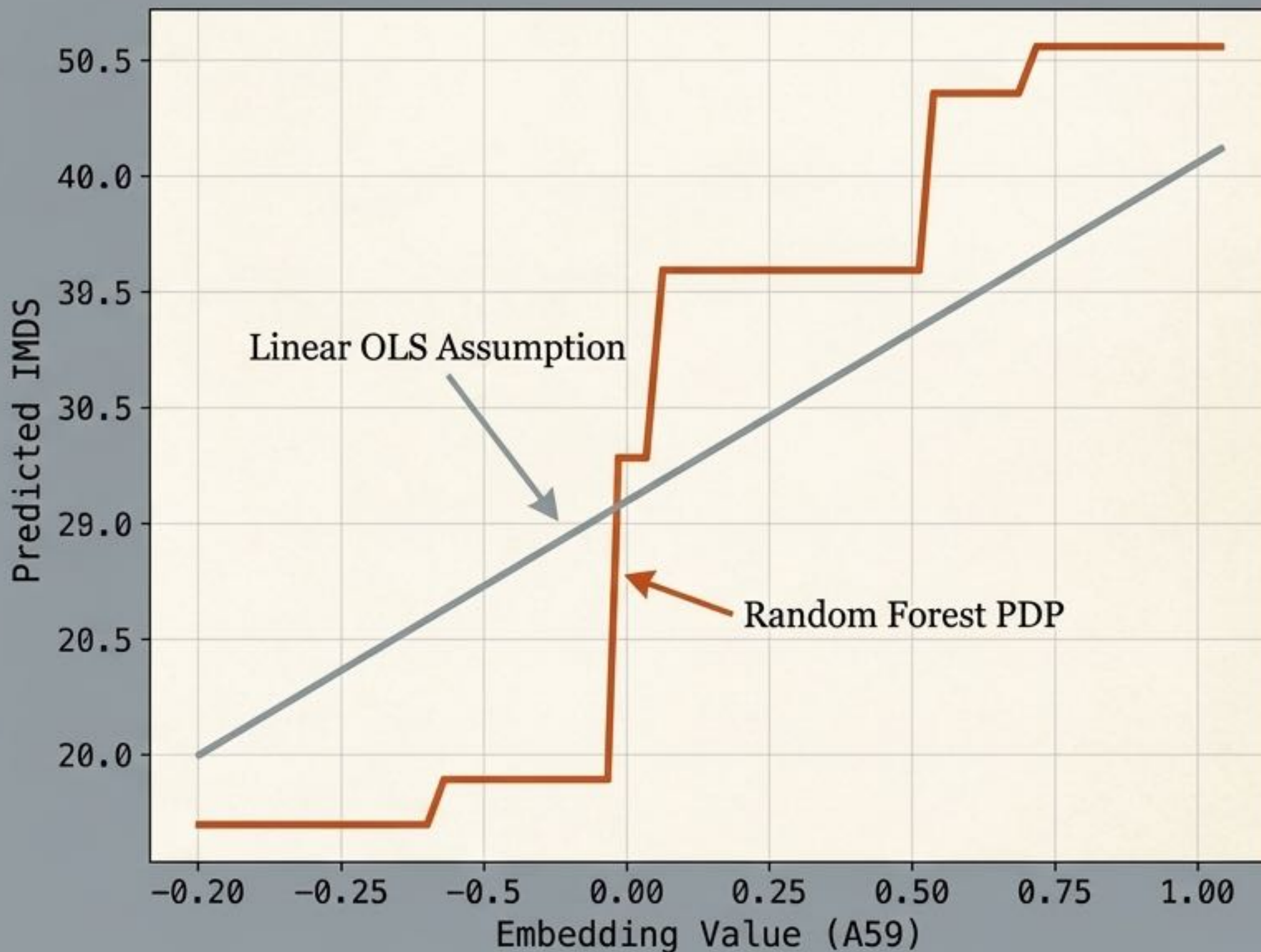
Partial Dependence: $f(x_k)$

Definition: The marginal effect of one feature on the prediction, averaging out all other features.

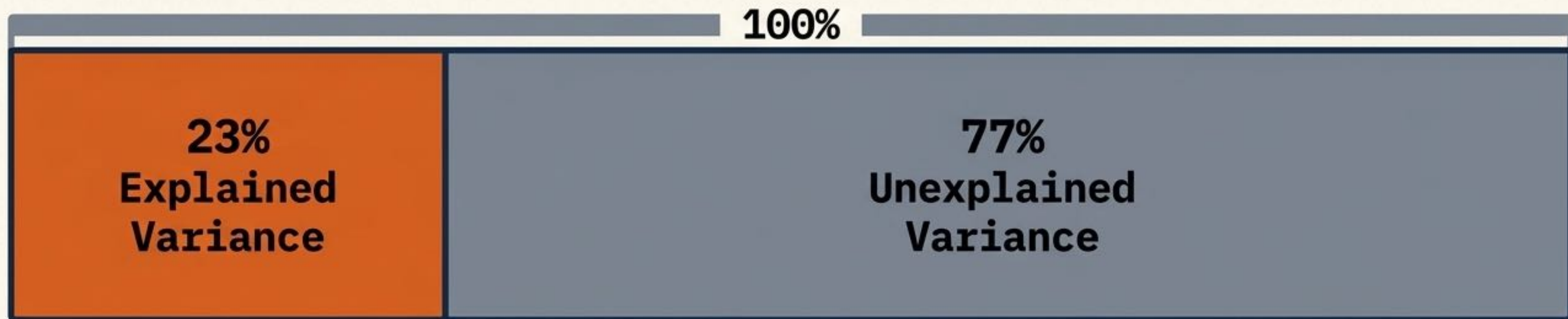
Analogy: Sliding the salt dial while keeping all other ingredients in the recipe perfectly fixed.

Diagnostic Insight: Threshold Effects

The plot uncovers sharp step-changes in predicted IMDS at specific spatial embedding values. Linear regression would entirely miss these abrupt phase transitions, completely justifying the use of tree-based methods over OLS for satellite data.



Economic Implications & Unexplained Variance



The Signal

Satellite embeddings capture a genuine, quantifiable macro-level development signal. Remote sensing is valid for baseline infrastructural proxies.

The Noise & Omitted Variable Bias

The vast majority of variation is driven by economic factors completely invisible from space:



Institutional
Governance Quality



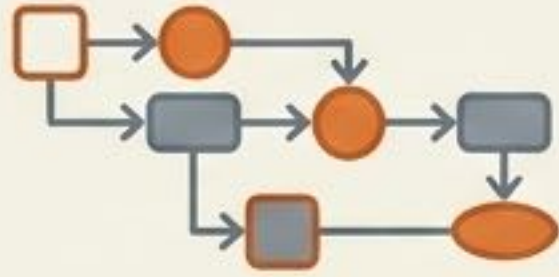
Informal
Economies



Migration &
Human Capital

Synthesis Conclusion: Satellite data is a powerful complement to, but never a substitute for, rigorous administrative survey data.

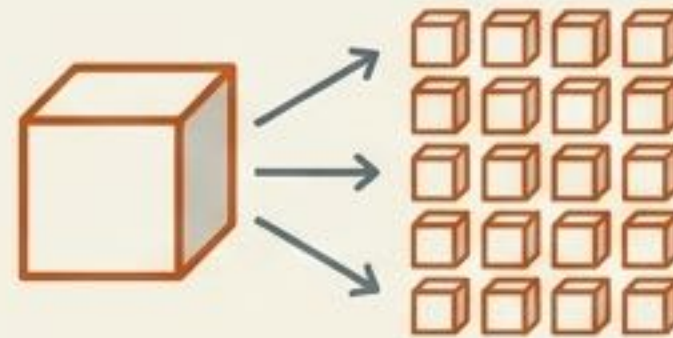
Methodological Extensions (Next Steps)



1. Algorithmic Iteration

Action: Implement GradientBoostingRegressor.

Hypothesis: Test if sequential error-correction algorithms yield improvements over parallel bagging for this specific spatial data structure.



2. Target Expansion

Action: Deconstruct the composite IMDS; predict specific, individual sub-indices (e.g., SDG1 vs SDG15).

Hypothesis: Isolate exactly which specific development vectors (e.g., infrastructure vs. education) are most visually apparent to satellites.

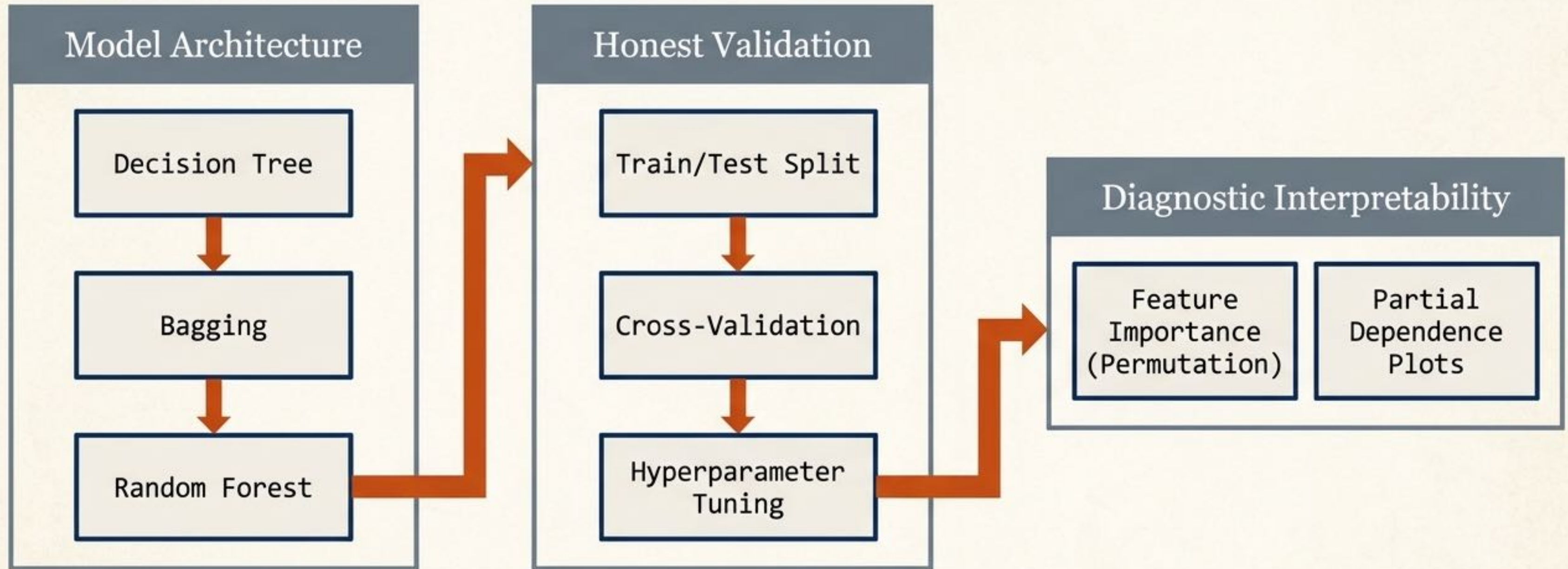


3. Spatial Augmentation

Action: Integrate administrative dummy variables at the departmental level.

Hypothesis: Capture unobserved regional fixed effects to absorb geographic variance that satellites naturally miss.

The Study Guide: Concept Synthesis Map



This methodological blueprint serves as the **baseline architecture** for applying predictive machine learning to spatial econometric challenges.